

PART 1 OF 2

TUTORIAL ON ATMEGA32 DEVELOPMENT BOARD



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Microcontrollers find applications in various industrial and domestic controls and processes. Therefore a good knowledge about them has become a necessity for electronics engineers. Although a lot of theoretical knowledge related to the software and hardware aspects of microcontrollers is available in the market, there is no easy alternative to a hands-on learning experience, and for that, a microcontroller development board is a must.

Here we describe a development board that is based on the popular AVR ATmega32 microcontroller and meets the needs of both the beginners as well as professionals. Beginners may use the board to learn the basics of microcontrollers, while professionals may use it for developing microcontroller-based complex and smart circuit designs. Twelve sample codes given in this project will also help one to learn more about the inbuilt peripherals of the microcontrollers.

The twelve program codes are Blink_LED, Chaser_Effect, Moving_Display, ADC_test, 8bit_Timer, Switches, External_INT, EEPROM, Freq_meter, PWM, UART and TWI.

Why AVR ATmega32?

The AVR is fast, low-cost, in-circuit programmable, and the development software is usually available for free as most of them are Open Source. Moreover, discussions and tutorials on the AVR family of processors are available on the Internet for solutions to development issues.

The ATmega32 is a high-perfor-

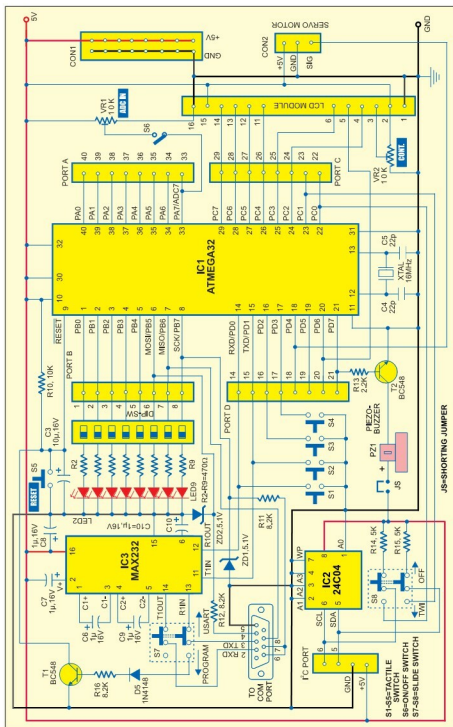


Fig. 1: Circuit of ATmega32 development board